

IF-ELSE IN PYTHON



PYTHON



IF STATEMENT IN PYTHON

An if statement in Python allows you to make decisions in your code based on certain conditions. It's like asking a question, and if the answer is "yes" (**True**), you do something; if the answer is "no" (**False**), you do something else or nothing at all.

Here's how to write an if statement in Python with a simple syntax:



```
if condition:
    # Statements to execute if
    # condition is true
```

Here, the condition after evaluation will be **either true or false**. if the statement accepts boolean values – if the value is true then it will execute the block of statements below it otherwise not.



INDENTATION (:

Indentation in Python is used to define code block. It **means adding spaces at the beginning of a line**. Proper indentation is crucial because Python uses it to understand the structure and flow of your code.

For Example:



```
if True:
    print("This is indented.")
    # This line is indented with 4 spaces
    print("This is part of the if block.")
    # This line is also indented

print("This is not indented.")
# This line is not part of the if block
```



KEY POINTS

- **Indentation Level:** Each block of code must be indented at the same level.
- **Consistency:** Typically, 4 spaces per indentation level are used.
- **Blocks:** Indentation is used after statements like if, for, while, functions, and classes to indicate the block of code that belongs to them.

Incorrect indentation will cause **errors in your code**. Here's an example of incorrect indentation:



```
if True:
print("This will cause an error.")
# This line is not indented
```

Make sure to indent properly to keep your code organized and error-free!



REAL-LIFE EXAMPLE

Imagine you have a weather app that tells you if you should bring an umbrella.



```
weather = "rainy"

if weather == "rainy":
    print("Bring an umbrella!")
else:
    print("No need for an umbrella.")
```

Here:

- The if statement checks if weather is "rainy".
- If weather is "rainy", it prints "Bring an umbrella!".
- If weather is not "rainy", it prints "No need for an umbrella." (handled by the else part).



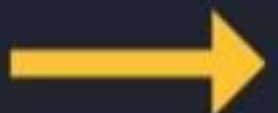
IF ELSE STATEMENT

An if-else statement is used in Python to execute certain code block based on conditions. It's a way for your program to make decisions and act accordingly.

if Statement Checks a condition. If the condition is **True**, it executes the code block inside the if. **else Statement Executes if the condition in the if statement is False.**

Basic Syntax

```
● ● ●  
if condition:  
    # code to execute if condition is True  
else:  
    # code to execute if condition is False
```



REAL-LIFE EXAMPLE

Imagine you are deciding what to wear based on the weather:

- **Condition:** If it is cold outside, you wear a jacket.
- **Else:** If it is not cold, you wear a t-shirt.

```
temperature = 10

if temperature < 15:
    print("Wear a jacket!")
else:
    print("Wear a t-shirt!")
```

Here:

- **Condition:** `temperature < 15` checks if the temperature is less than 15 degrees.
- **if Block:** If the condition is True (temperature is less than 15 degrees), it prints "Wear a jacket!".
- **else Block:** If the condition is False (temperature is 15 degrees or higher), it prints "Wear a t-shirt!".



IF-ELIF-ELSE STATEMENT

An if-elif-else statement allows you to **check multiple conditions in sequence**. It is useful when you have more than two possible outcomes and need to choose one based on different conditions.

- **if Statement:** Checks the first condition. If it's True, executes the corresponding block of code.
- **elif Statement:** Short for "else if". Checks another condition if the previous if or elif conditions were False. You can have multiple elif statements.
- **else Statement:** Executes if none of the if or elif conditions are True.



BASIC SYNTAX

```
if condition1:
    # code to execute if condition1 is True
elif condition2:
    # code to execute if condition2 is True
else:
    # code to execute if none of the above are True
```



REAL-LIFE EXAMPLE

Imagine you are deciding what to wear based on the temperature:

- **Condition 1:** If it is very cold, you wear a coat.
- **Condition 2:** If it is moderately cold, you wear a jacket.
- **Else:** If it is warm, you wear a t-shirt.

```
● ● ●
temperature = 10

if temperature < 5:
    print("Wear a coat!")
elif temperature < 15:
    print("Wear a jacket!")
else:
    print("Wear a t-shirt!")
```

